

Deep Fake Detection With EfficientNet and FasterViT

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Abstract— This report presents a comprehensive study on deep fake detection using EfficientNet and FasterViT. Our approach leverages the efficiency and accuracy of EfficientNet, combined with the speed and scalability of FasterViT, to detect manipulated images. The project demonstrates significant success in identifying deep fakes, contributing to the field of digital media verification.

Keywords— Deep Fake Detection, Vision Transformers, EfficientNet, FasterViT, Machine Learning, GradCAM, Vision Image Transformer

I. INTRODUCTION

This project focuses on leveraging the capabilities of vision transformers for the detection of deep fakes. Deepfakes, synthetic media in which a person's likeness is replaced with someone else's, pose significant challenges in the realms of security, privacy, and information integrity. Our goal is to develop a robust model capable of accurately distinguishing between real and deep fake images by using the latest advancements in machine learning and image processing.

The core problem addressed in this project is the detection of deep fake images. Deepfakes, created using sophisticated AI techniques, are increasingly realistic and challenging to identify. The primary objective is to train a vision transformer model that can effectively discern between authentic images and deepfakes with high accuracy. This problem is significant due to the rising prevalence of deepfake technology and its potential misuse in various domains, including politics, entertainment, and social media.

II. METHODS

Our methodology involved adapting the EfficientNet-B3 model for binary classification and experimenting with FasterViT to compare performance. We implemented data augmentation techniques, including random flips and color

adjustments, to enhance the model's generalization capabilities. Dropout and L2 regularization were used to prevent overfitting.

$$\text{Dropout Regularization: } \sum_{i=1}^n \delta_i w_i^t I_i$$

$$\delta = \text{Bernoulli}(p)$$

$$P(n) = \begin{cases} 1-p & \text{for } n = 0 \\ p & \text{for } n = 1, \end{cases}$$

$$\text{L2 Regularization: } \lambda \sum_{i=1}^k w_i^2$$

Random Erasing is a data augmentation method that involves randomly selecting and erasing a rectangular region in an input image during training. This technique helps to improve the robustness and generalization of deep neural networks by introducing variations in the training data. In our case, we made usage of Random Erasing for background removal in order to fix issues with our earlier models focusing on the backgrounds for real or fake determination.



Fig. 1 Fake Obama image, before usage of RandomErasing.

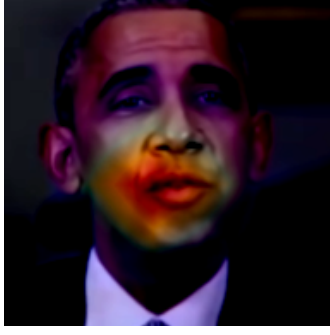


Fig. 2 Fake Obama image, after using RandomErasing and other model improvements.

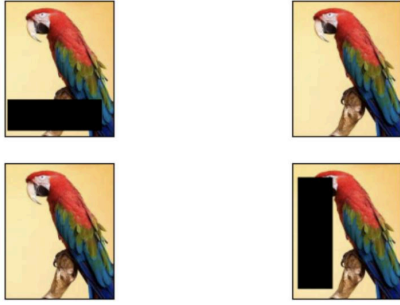


Fig. 3 Example images of RandomErasing

In addition to RandomErasing we also implemented some random image augmentations already made available via the torchvision library such as; RandomHorizontalFlip, RandomRotate, and ColorJitter.

III. EXPERIMENTS

Our experimental phase was extensive, involving trials with various architectures and datasets to determine the most effective approach for deep fake detection. Initially, we experimented with the CIFAR-10 dataset to establish a baseline for model performance. While CIFAR-10 provided a good starting point, its simplicity compared to the complexity of deep fake detection necessitated a transition to more challenging datasets.

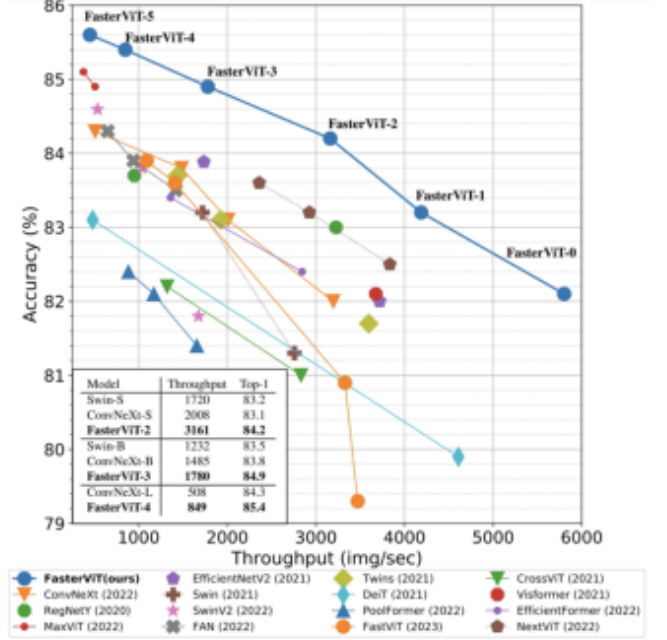


Fig. 4 Throughput Speed VS Accuracy of leading image classification models.

We explored several neural network architectures, including Swin Transformer V2 (SwinV2) and ResNet. SwinV2, known for its effectiveness in various vision tasks, was initially promising due to its advanced architecture. However, we encountered challenges with its processing speed, which was a critical factor given the large volume of data in deep fake detection. ResNet, another popular choice for image classification tasks, was tested for its renowned efficiency and performance. Despite its strengths, ResNet did not achieve the level of accuracy we aimed for in distinguishing between real and manipulated images.

Ultimately, our experiments led us to focus on EfficientNet due to its balance of efficiency and accuracy. The model's performance on our chosen dataset, which included high-resolution images of human faces categorized as real or fake, was impressive. We also experimented with FasterViT to assess its potential in enhancing detection speed and scalability, which showed promising results, though it required further tuning for optimal accuracy.

IV. RESULTS

The EfficientNet-B model achieved a test accuracy of 87.06%, demonstrating its effectiveness in classifying images. The confusion matrix provided (figure 5) insights into the model's classification behavior. We recognized during our testing stages that EfficientNet-B was incapable of determining a large number of real images.

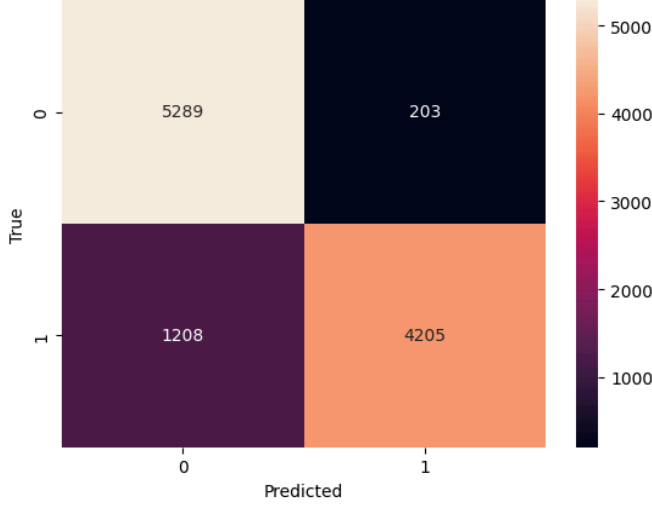


Fig. 5 Confusion Matrix of EfficientNet-B model at 3 epochs. But it was found to be overfitting

In order to improve our accuracy we implemented FasterViT, a vision transformer developed for image classification. We decided to run a trial on 1 epoch with FasterViT-0 and got an accuracy score of 88.92%, its confusion matrix has been added (figure 6). Some problems with FasterViT-0 were that we had issues of overfitting, in order to fix said problems, we switched to FasterViT-2 at 3 epochs. Implementation of the FasterViT-2 proved extremely promising, our first score surpassing EfficientNet achieved 89.96% accuracy.

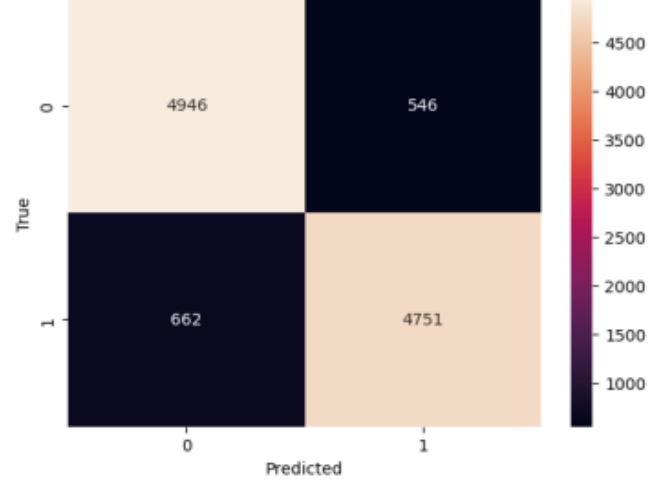


Fig. 6 Confusion Matrix of FasterViT-0 model at 1 epoch.

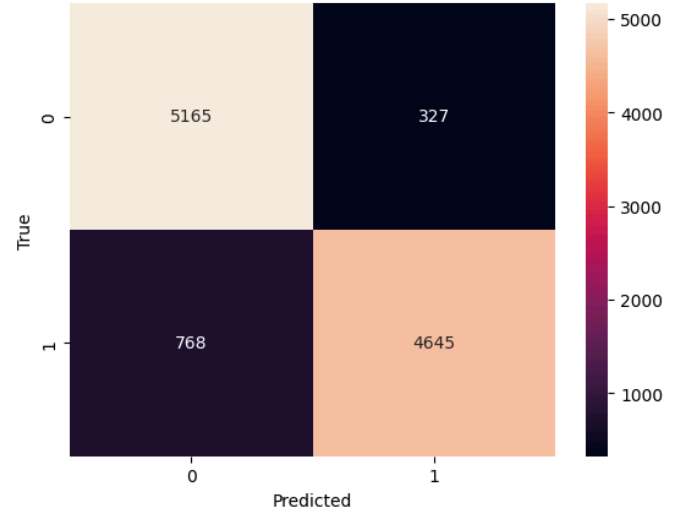


Fig. 7 Confusion Matrix of FasterViT-2 model at 3 epochs. Similar to our 3 epoch EfficientNet, it still struggles at determining real images.

V. RELATED WORK

The field of deep fake detection has seen significant advancements in recent years, with various approaches being explored to tackle this growing challenge. A notable contribution in this domain is the FaceForensics++ dataset, introduced in 2019. This dataset has been instrumental in advancing deep fake detection research, providing a large collection of manipulated videos to facilitate the development and testing of detection algorithms. Researchers have used FaceForensics++ to train models on distinguishing between real and altered facial features in videos, employing

techniques ranging from traditional machine learning to more advanced deep learning methods. The dataset's diversity and complexity have made it a benchmark for assessing the performance of deep fake detection models, highlighting the importance of comprehensive and varied data in training effective detection systems.

In parallel to the development of datasets like FaceForensics++, there has been a surge in exploring various neural network architectures for deep fake detection. Prior studies have predominantly focused on convolutional neural networks (CNNs) due to their proven efficacy in image and video recognition tasks. The introduction of architectures like EfficientNet, which offers a balanced scale of depth, width, and resolution, has opened new avenues in this field. EfficientNet's ability to achieve high accuracy with relatively fewer parameters makes it an attractive option for deep fake detection. More recently, the exploration of transformer-based models like FasterViT has added a new dimension to this research area. FasterViT's emphasis on speed and scalability addresses the growing need for efficient processing of large volumes of data, a common challenge in deep fake detection. These developments reflect the dynamic nature of the field, where continuous innovation is key to staying ahead of increasingly sophisticated deep fake technologies.

VI. CONCLUSION

This project represents a significant step forward in the field of deep fake detection. Through our extensive experimentation with various datasets and neural network architectures, we have gained valuable insights into the challenges and requirements of effective deep fake detection. Our findings underscore the importance of choosing the right model architecture that balances accuracy, efficiency, and processing speed. EfficientNet emerged as the most suitable model for our needs, demonstrating high accuracy in identifying deep fakes. The exploration of FasterViT also highlighted the potential benefits of transformer-based models in handling large-scale data efficiently.

The journey through this project has not only advanced our understanding of deep fake detection but also contributed to the broader field of digital media verification. As deep fakes become increasingly sophisticated, the need for robust detection methods will continue to grow. Our work lays the groundwork for future research in this area, emphasizing the need for continuous innovation and adaptation in response to evolving deep fake technologies. The lessons learned from experimenting with different models and datasets will inform future efforts to develop more advanced and reliable deep fake detection systems.

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